Example 4 *The exact integral of* $\sin(x)$ *over* $[0, \pi]$ *equals two. Computing* $\tilde{u}_h$ *with the trapezoidal rule and plotting* $|\tilde{u}_h - 2|$ *in a* `loglog` *plot we get the result shown in Figure 1.*

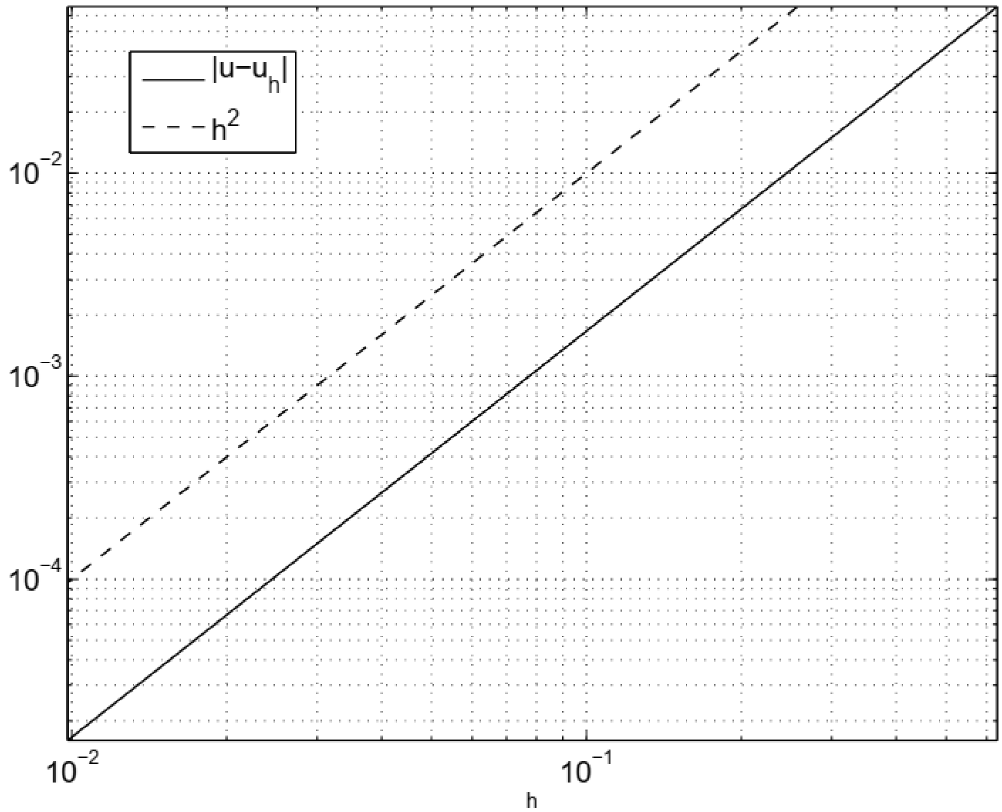

Figure 1. Error in trapezoidal rule for $f(x) = \sin(x)$ as a function of h. The dashed line is h^2 as a function of h which has precisely slope two. It thus indicates the slope for a second order method, for comparison.

h	$\tilde{u}_h$	$\tilde{u}_h - \tilde{u}_{h/2}$	$\frac{\tilde{u}_h - \tilde{u}_{h/2}}{\tilde{u}_{h/2} - \tilde{u}_{h/4}}$	$\log_2 \frac{\tilde{u}_h - \tilde{u}_{h/2}}{\tilde{u}_{h/2} - \tilde{u}_{h/4}}$
$\pi/5$	1.933765598092805	-0.049757939416650	4.024930251575880	2.008963782835339
$\pi/10$	1.983523537509455	-0.012362435199260	4.006184396966857	2.002228827158397
$\pi/20$	1.995885972708715	-0.003085837788350	4.001543117204195	2.000556454557076
$\pi/40$	1.998971810497066	-0.000771161948770	4.000385593360853	2.000139066704584
$\pi/80$	1.999742972445836	-0.000192771904301	4.000096386716427	2.000034763740606
$\pi/160$	1.999935744350136	-0.000048191814813		
$\pi/320$	1.999983936164949			

Table 1. Table of values for the trapezoidal rule for $f(x) = \sin(x)$. The last column is the final approximation of the order of accuracy p.